

# Variational Inference for Bayesian Computation

An Application to NBA Salary Data

## Background

In Bayesian statistics, the main object of interest is the posterior distribution

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})}, \quad (1)$$

which combines information from the observed data through the likelihood  $p(\mathbf{y} \mid \boldsymbol{\theta})$  with prior information through  $p(\boldsymbol{\theta})$ ; see Gelman et al. (2013). For many models, however, the marginal likelihood  $p(\mathbf{y})$  is not available in closed form. Consequently, the posterior distribution cannot be computed analytically and approximate inference methods are required.

Two important approaches are Markov Chain Monte Carlo (MCMC) and Variational Bayes (VB). MCMC methods generate samples from the posterior distribution and can provide highly accurate approximations if the chain is run long enough (Casella and George, 1992). However, this accuracy often comes at a high computational cost, especially for large datasets or models with many parameters.

Variational Bayes takes a different approach. Instead of sampling from the posterior, it chooses a simpler family of distributions  $q(\boldsymbol{\theta})$  and searches for the member of this family that minimizes the Kullback–Leibler (KL) divergence to the true posterior distribution (Ormerod and Wand, 2010). This transforms Variational Bayes into an optimization problem. As a result, VB is usually much faster than MCMC, but the approximation quality depends on the chosen variational family.

## Mean-Field VB in Action

To make Variational Bayes concrete, we consider Bayesian linear regression. The model is given by

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_n), \quad (2)$$

where  $\mathbf{y} \in \mathbb{R}^n$  is a vector of observations from the response variable,  $X \in \mathbb{R}^{n \times p}$  is the design matrix, and  $\boldsymbol{\beta} \in \mathbb{R}^p$  is the vector of regression coefficients. Hence, the likelihood is proportional to the Gaussian density:

$$p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) \propto (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}) \right\}. \quad (3)$$

We assume independent priors

$$\boldsymbol{\beta} \sim \mathcal{N}_p(\boldsymbol{\mu}_\beta, \Sigma_\beta), \quad \sigma^2 \sim \text{IG}(A, B). \quad (4)$$

These priors are standard choices in Bayesian linear regression. They are convenient because they lead to tractable posterior expressions and closed-form coordinate updates in the variational approximation (Gelman et al., 2013).

For the model specified above, the target distribution is the joint posterior

$$p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, X) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2, X) p(\boldsymbol{\beta}) p(\sigma^2). \quad (5)$$

Instead of approximating this posterior by samples, Variational Bayes replaces it by a simpler distribution  $q(\boldsymbol{\beta}, \sigma^2)$

and optimizes this approximation. We use the mean-field factorization

$$q(\boldsymbol{\beta}, \sigma^2) = q(\boldsymbol{\beta}) q(\sigma^2). \quad (6)$$

This factorization assumes independence between  $\boldsymbol{\beta}$  and  $\sigma^2$  under the variational distribution. The two variational factors are obtained from the expected log joint density (Ormerod and Wand, 2010):

$$\log q(\boldsymbol{\beta}) \propto \mathbb{E}_{q(\sigma^2)} [\log p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) + \log p(\boldsymbol{\beta})], \quad (7)$$

$$\log q(\sigma^2) \propto \mathbb{E}_{q(\boldsymbol{\beta})} [\log p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) + \log p(\sigma^2)]. \quad (8)$$

Each variational factor can be computed analytically by first evaluating the corresponding expectation and then recognizing the kernel of a known distribution. Specifically, the update for  $q(\boldsymbol{\beta})$  has the kernel of a multivariate normal distribution, while the update for  $q(\sigma^2)$  has the kernel of an inverse-gamma distribution. Hence,

$$\begin{aligned} q(\boldsymbol{\beta}) &= \mathcal{N}_p(\boldsymbol{\mu}_{q(\boldsymbol{\beta})}, \Sigma_{q(\boldsymbol{\beta})}), \\ q(\sigma^2) &= \text{IG}(A_{q(\sigma^2)}, B_{q(\sigma^2)}), \end{aligned} \quad (9)$$

where the variational parameters are equal to:

$$A_{q(\sigma^2)} = A + \frac{n}{2}, \quad (10)$$

$$\begin{aligned} B_{q(\sigma^2)} &= B + \frac{1}{2} \left[ (\mathbf{y} - X\boldsymbol{\mu}_{q(\boldsymbol{\beta})})^\top (\mathbf{y} - X\boldsymbol{\mu}_{q(\boldsymbol{\beta})}) \right. \\ &\quad \left. + \text{tr}(X^\top X \Sigma_{q(\boldsymbol{\beta})}) \right], \end{aligned} \quad (11)$$

$$\Sigma_{q(\boldsymbol{\beta})} = \left( \frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}} X^\top X + \Sigma_\beta^{-1} \right)^{-1}, \quad (12)$$

$$\boldsymbol{\mu}_{q(\boldsymbol{\beta})} = \Sigma_{q(\boldsymbol{\beta})} \left( \frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}} X^\top \mathbf{y} + \Sigma_\beta^{-1} \boldsymbol{\mu}_\beta \right). \quad (13)$$

In the CAVI algorithm, the parameters are updated iteratively:  $\Sigma_{q(\boldsymbol{\beta})}$  and  $\boldsymbol{\mu}_{q(\boldsymbol{\beta})}$  depend on  $A_{q(\sigma^2)}$  and  $B_{q(\sigma^2)}$ , while  $B_{q(\sigma^2)}$  depends on  $\boldsymbol{\mu}_{q(\boldsymbol{\beta})}$  and  $\Sigma_{q(\boldsymbol{\beta})}$ .

The CAVI updates can be understood as coordinate-wise maximization of the Evidence Lower Bound (ELBO) (Ormerod and Wand, 2010). Using the mean-field factorization from Equation (6), the ELBO is

$$\begin{aligned} \mathcal{L}(q) &= \underbrace{\mathbb{E}_q[\log p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2)]}_{\text{Likelihood}} + \underbrace{\mathbb{E}_q[\log p(\boldsymbol{\beta})]}_{\boldsymbol{\beta}\text{-Prior}} + \underbrace{\mathbb{E}_q[\log p(\sigma^2)]}_{\sigma^2\text{-Prior}} \\ &\quad - \underbrace{\mathbb{E}_q[\log q(\boldsymbol{\beta})]}_{\boldsymbol{\beta}\text{-Entropy}} - \underbrace{\mathbb{E}_q[\log q(\sigma^2)]}_{\sigma^2\text{-Entropy}}. \end{aligned} \quad (14)$$

In the model considered here, the ELBO can be computed

analytically in closed form as:

$$\begin{aligned}
\mathcal{L}(q) = & -\frac{n}{2} \log(2\pi) - \frac{n}{2} \left\{ \log(B_{q(\sigma^2)}) - \psi(A_{q(\sigma^2)}) \right\} \\
& - \frac{1}{2} \frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}} \left[ (\mathbf{y} - X\boldsymbol{\mu}_{q(\beta)})^\top \right. \\
& \quad \left. (\mathbf{y} - X\boldsymbol{\mu}_{q(\beta)}) + \text{tr}(X^\top X \Sigma_{q(\beta)}) \right] \\
& - \frac{p}{2} \log(2\pi) - \frac{1}{2} \log|\Sigma_\beta| \\
& - \frac{1}{2} \left[ \text{tr}(\Sigma_\beta^{-1} \Sigma_{q(\beta)}) \right. \\
& \quad \left. + (\boldsymbol{\mu}_{q(\beta)} - \boldsymbol{\mu}_\beta)^\top \Sigma_\beta^{-1} (\boldsymbol{\mu}_{q(\beta)} - \boldsymbol{\mu}_\beta) \right] \\
& + A \log B - \log \Gamma(A) \\
& - (A+1) \left\{ \log(B_{q(\sigma^2)}) - \psi(A_{q(\sigma^2)}) \right\} \\
& - B \frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}} + \frac{1}{2} \log|\Sigma_{q(\beta)}| \\
& + \frac{p}{2} \{1 + \log(2\pi)\} + A_{q(\sigma^2)} + \log(B_{q(\sigma^2)}) \\
& + \log \Gamma(A_{q(\sigma^2)}) - (A_{q(\sigma^2)} + 1) \psi(A_{q(\sigma^2)}). \quad (15)
\end{aligned}$$

During CAVI, the data  $X, \mathbf{y}$  and the prior hyperparameters  $\boldsymbol{\mu}_\beta, \Sigma_\beta, A, B$  remain fixed. Only the variational parameters  $\boldsymbol{\mu}_{q(\beta)}, \Sigma_{q(\beta)}, A_{q(\sigma^2)}$ , and  $B_{q(\sigma^2)}$  are updated iteratively until convergence of the ELBO.

## Simulation Study

To illustrate the computational motivation for Variational Bayes, we compare CAVI and Gibbs sampling in the Bayesian linear regression model from Equation (2). We simulate data for increasing problem sizes  $(n, p)$ , while keeping the ratio  $p/n$  fixed. Here,  $p$  denotes the number of regression coefficients including the intercept.

Table 1: Runtime comparison.

| $n$   | $p$  | CAVI time (s) | Gibbs time (s) |
|-------|------|---------------|----------------|
| 100   | 10   | < 0.01        | 0.19           |
| 1000  | 100  | 0.14          | 1.70           |
| 5000  | 500  | 16.05         | 151.15         |
| 10000 | 1000 | 128.53        | 1414.59        |

Table 1 shows that the runtime gap becomes more pronounced as the problem size increases. In the largest setting, CAVI requires about 129 seconds, whereas Gibbs sampling takes about 1415 seconds, or roughly 23.6 minutes, for 1000 iterations.

## Application to NBA Salary Data

We apply the Bayesian linear regression model from Equation (2) to NBA salary data (Ratin21, n.d.). The response is standardized log salary, and  $X$  contains standardized season and player-performance variables such as age, games played, shooting statistics, rebounds, assists, blocks, fouls and points. The final data set contains  $n = 7298$  player-seasons and  $p = 19$  coefficients including the intercept. We compare mean-field VB with Gibbs sampling.

MFVB and MCMC give almost identical posterior means and intervals; we interpret coefficients whose 95% posterior intervals exclude zero. Since the model was estimated on standardized variables, the selected coefficients are transformed back to the original log-salary scale by

$$\beta_j^{\text{orig}} = \beta_j^{\text{scaled}} \frac{s_y}{s_{x_j}}, \quad j = 1, \dots, p-1, \quad (16)$$

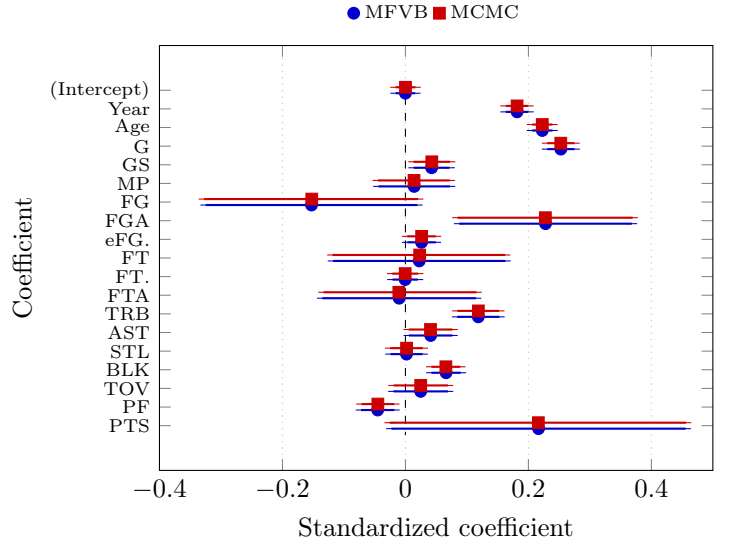


Figure 1: Comparison of MFVB and MCMC. Error bars show 95% posterior intervals.

Table 2: Selected effects on salary.

| Variable | Unit change         | MFVB (%) | MCMC (%) |
|----------|---------------------|----------|----------|
| Year     | +1 raw unit         | 5.74     | 5.74     |
| Age      | +1 raw unit         | 7.74     | 7.74     |
| G        | +1 raw unit         | 1.52     | 1.52     |
| GS       | +1 raw unit         | 0.22     | 0.22     |
| FGA      | +1 raw unit         | 7.23     | 7.22     |
| eFGperc  | +1 percentage point | 0.39     | 0.39     |
| TRB      | +1 raw unit         | 7.18     | 7.18     |
| AST      | +1 raw unit         | 3.22     | 3.20     |
| BLK      | +1 raw unit         | 24.72    | 24.59    |
| PF       | +1 raw unit         | -8.20    | -8.14    |

and percentage effects are computed as

$$100 \left( \exp(\beta_j^{\text{orig}} \Delta x_j) - 1 \right) \%. \quad (17)$$

All effects are interpreted while holding the remaining predictors constant. For example, one additional year is associated with about 5.74% higher salary, and one additional year of age with about 7.74% higher salary. Performance variables such as field goal attempts, rebounds, assists and blocks show positive effects, while personal fouls are negatively associated with salary.

## Conclusion

Mean-field Variational Bayes provides a deterministic and computationally efficient alternative to MCMC. In the simulation study, CAVI is substantially faster than Gibbs sampling, especially for larger problem sizes. In the NBA salary application, MFVB and MCMC lead to almost identical posterior coefficient estimates. Possible extensions include richer variational families or stochastic variational inference for larger data sets.

## References

- Casella, G. and George, E. I. (1992). Explaining the Gibbs Sampler. *The American Statistician*.
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC.
- Ormerod, J. T. and Wand, M. P. (2010). Explaining variational approximations. *The American Statistician*.
- Ratin21. (n.d.). NBA Player Stats and Salaries 2010–2025. Kaggle. <https://www.kaggle.com/datasets/ratin21/nba-player-stats-and-salaries-2010-2025>